

MANISH BIKUMALLA

+1(716)-465-9794 | NY, USA | [linkedin.com/in/manishbikumalla](https://www.linkedin.com/in/manishbikumalla) | manishbi@buffalo.edu

Software Developer with over 2 years of experience with a strong foundation in software development principles, data structures, and algorithms, with hands-on experience in building distributed applications using Java, Python and AWS. Skilled in implementing machine learning models and committed to developing high-performance applications. Eager to contribute to challenging, real-world projects and drive impactful solutions.

Technical Skills

Languages & Web Technologies: Python, Java, HTML, CSS, SQL

Libraries & Frameworks: Numpy, Pandas, Streamlit, Pytorch, Flask, Hadoop, Apache Spark, Spring Boot, Hibernate

APIs & Databases: REST APIs, MySQL, MongoDB

Tools & Version Control: Docker, VMware, Microsoft Office Suite, Git, GitHub, Agile, Maven, AWS, Snowflake, Kafka

Coursework – Algorithms, Analysis and Design, Data Models and Query Languages, Deep Learning, Data Intensive Computing, Operating systems, Computer Vision

Education

University at Buffalo, State University of New York

January 2024 – May 2025 (Expected)

Master of Science in Computer Science and Engineering (GPA – 3.7/4)

IIT Kharagpur, India

B.Tech in Metallurgical and Materials Engineering (GPA – 3.6/4)

Experience

Software Developer, CBIC

November 2021 – January 2024

- Engineered a suite of **Java and Spring Boot** based **REST APIs** for tax evasion detection which resulted in up to 24% indirect tax revenue for the Financial Year 2022-23.
- Streamlined processing of thousands of tax returns by redesigning database schema and backend logic for the GSTR9A tax form.
- Automated monthly brochure generation through a **dedicated microservice**, cutting production time by at least 40%.
- Built a real-time notification system using **Spring validation and Java**, alerting taxpayers about deficits in return filing errors, this led to reduced manual communication workload by 20% and improved compliance.
- Designed and executed a large-scale data processing pipeline from application data source **MongoDB** to **Snowflake** using **Apache Spark (Java)**, enabling real-time access to critical insights for over 2,000 employees via the Advait portal.
- Implemented and maintained **CI/CD pipelines** with Jenkins and Docker reducing deployment time by 15%, enabling consistent and efficient software releases.
- Collaborated with cross functional teams in an Agile environment, practiced test driven development (TDD) and actively integrated stakeholder feedback to improve project outcomes.

Projects

Chirp – a chatting app | *Java, AWS, MySQL*

- Built a distributed chat application using **Java and Springboot**, facilitating peer-to-peer communication among users through an intuitive and user-friendly interface.
- Devised a robust message queuing system on AWS with SQS, load balancing and rate limiting to ensure reliable performance and system stability.
- Utilized MySQL to reliably store and manage message data, complementing the AWS SQS queue for persistent storage.

ChestAI | *Python, Pytorch, Deep Learning*

- Created** a deep learning model using Streamlit and Pytorch (ResNet and DenseNet architectures) for chest X-ray classification and **achieved 92.85% accuracy**.
- Enhanced model performance and computational efficiency by integrating GANs for data augmentation and autoencoders for dimensionality reduction, resulting in a 90% reduction in training time and improved diagnostic accuracy.

OurHealthAI | *LLMs, HTML, CSS*

- Developed a healthcare platform with features like a chatbot for answering insurance-related queries, secure user authentication, and a responsive design.
- Implemented a RAG model using Langchain** and PyPDF for PDF data parsing to enable the chatbot to provide accurate responses based on pre-fed insurance data.

TransferPredictor | *Python, Streamlit, Numpy, Pandas, Machine Learning*

- Architected and executed a comprehensive machine learning pipeline using XGBoost and Random Forest to predict football players' market value and wage rates.
- Deployed the model as an interactive web application using **Streamlit**, enabling real-time predictions and insightful visualizations for decision-making in player recruitment and management.